Ruei-Che Chang

Curriculum Vitae

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EDUCATION

2020– **National Taiwan University**, *Taipei, Taiwan*.
Visiting Student at Interactive Graphics Lab.
Advisor: Bing-Yu Chen.

2019–2021 **Dartmouth College**, *New Hampshire, Hanover*.
Master of Science in Computer Science. *75% Tuition Scholarship from CS department*.
Advisor: Xing-Dong Yang.
Thesis: AccessibleCircuits: Adaptive Add-on Circuit Components for People with Blindness or Low Vision.

2014–2018 **National Cheng Kung University**, *Tainan Taiwan*.
Bachelor of Science in Electrical Engineering.

PUBLICATIONS

2021 [C.6] **Ruei-Che Chang**, Chih-An Tsao, Fang-Ying Liao, Seraphina Yong, Tom Yeh, Bing-Yu Chen, "Daedalus in the Dark: Designing for Non-Visual Accessible Construction of Laser-Cut Architecture." *In Proceedings of the ACM Symposium on User Interface Software & Technology* (**UIST'21**).

[C.5] **Ruei-Che Chang**, Wen-Ping Wang, Chi-Huan Chiang, Justin Luo, Te-Yen Wu, Zheer Xu, Bing-Yu Chen, Xing-Dong Yang, "AccessibleCircuits: Adaptive Add-On Circuit Components for People with Blindness or Low Vision." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'21**).

2020 [C.4] **Ruei-Che Chang**, Chi-Huan Chiang, Shuo-wen Hsu, Chih-Yun Yang, Da-Yuan Huang, Bing-Yu Chen, "TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays." *In Proceedings of the ACM Conference on Spatial User Interface* (**SUI'20**).

[C.3] Yi-Shyuan Chiang*, **Ruei-Che Chang***, Yi-Lin Chuang, Shih-Ya Chou, Hao-Ping Lee, I-Ju Lin, Jian Hua Jiang Chen, Yung-Ju Chang, "Exploring the Design Space of User-System Communication for Smart home Routine Assistants." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'20**) (*co-primary).

[C.2] Kai-Chieh Huang, Chen-Kuo Sun, Da-Yuan Huang, Yu-Chun Chen, **Ruei-Che Chang**, Shuo-wen Hsu, Chih-Yun Yang, Bing-Yu Chen, "Glissade: Generating Balance Shifting Feedback to Facilitate Auxiliary Digital Pen Input." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'20**).

2019 [C.1] Chi Wang, Da-Yuan Huang, Shuo-Wen Hsu, Chu-En Hou, Yeu-Luen Chiu, **Ruei-Che Chang**, Jo-Yu Lo, Bing-Yu Chen, "Masque: Exploring Lateral Skin Stretch Feedback on the Face with Head-Mounted Displays." *In Proceedings of the ACM Symposium on User Interface Software & Technology* (**UIST'19**).

AWARDS

2020 **Honorable Mention Award for Glissade [C.2]** at ACM CHI 2020.

2019 Taiwan internship scholarship from the Ministry of Education (acceptance rate 21.53%).

2017 2nd award at Taiwan Intelligent Robot Contest (30 participants).

RESEARCH EXPERIENCE

Jun 2020 – present **Interactive Graphics Lab, National Taiwan University**
Visiting student, advised by Bing-Yu Chen and Tom Yeh (Visiting scholar).

- Designing and conducting a study to identify accessibility challenges of laser-cut assembly. Implementing a Fusion 360 plug-in to generate tactile aids for input laser-cut model. Conducting an evaluation for tactile aids, and leading video and qualitative analysis [C.6].
- Collecting brain signal data using fNIRs on VR tasks, and analyzing data by Matlab.
- Integrating arduino and electronics, and implementing two VR applications. Designing and conducting study [C.4].

Feb 2019 – Aug 2019 ***Research Assistant, advised by Bing-Yu Chen and Da-Yuan Huang.***
Using Unity to implement six tablet applications for [C.2], and an authoring tool for designer to specify haptic feedback on a tablet [C.1].

Sep 2019 – Mar 2021 **XDiscovery Lab, Dartmouth College**
Graduate student, advised by Xing-Dong Yang.

- Designing and conducting a study to identify accessibility challenges of circuit prototyping, and an evaluation for AccessibleCircuits [C.5], where I designed 3D-printed adaptions for existing circuit components via Fusion 360, and implementing a haptic-audio interface compatible to Android smartphone.
- Collecting and labeling 500+ images of breadboard circuits, and implementing a computer vision-based system using APIs (e.g., Mask-RCNN and OpenCV), and web interfaces (e.g., socket.io, MongoDB) on both student and teacher ends for remote learning.

Sep 2019 – Apr 2019 **Mobile and Ubiquitous Interaction Lab, National Yang Ming Chiao Tung University**
Research Assistant, advised by Yung-Ju (Stanley) Chang.

- Design and execute an experiment using methods, such as WoZ, speed dating, user enactment and contexual inquiry, to understand user-smart agent communication [C.3].

PRESENTATION

TAICHI'21 Invited to Best of HCI session to present *Daedalus*[*C.6*] and *AccessibleCircuits*[*C.5*] (TAICHI is the largest domestic HCI conference in Taiwan).

UIST'21 Daedalus in the Dark: Designing for Non-Visual Accessible Construction of Laser-Cut Architecture (Held Virtually).

CHI'20 AccessibleCircuits: Adaptive Add-On Circuit Components for People with Blindness or Low Vision (Held Virtually).

SUI'20 TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays (Held Virtually).

REVIEWER

UIST '21

CHI,LBW '20 '21

SKILLS

Programming Java, C#, Python, HTML, JavaScript, ROS, OpenCV, MongoDB, LaTeX

Prototyping Arduino, 3D-printing, Fusion 360, laser-cutting

Platforms Unity3D, Android Studio, Fusion 360